A Calibrated, Cost-Sensitive, and Explainable Machine Learning Framework for Stroke Risk Prediction on Imbalanced Clinical Data

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Research Domain: Explainable Artificial Intelligence (XAI) for Stroke Prediction
Dataset: Official Kaggle Healthcare Stroke Prediction Dataset (N = 5,110)

***Abstract*—Cerebrovascular accidents (strokes) represent the second leading cause of global mortality and the third leading contributor to long-term cognitive and functional disability. While supervised machine learning algorithms demonstrate competitive discriminative potential in identifying vascular vulnerability, clinical deployment is obstructed by three interconnected challenges: (1) extreme class imbalance (<5% positive prevalence in community screening cohorts), (2) miscalibrated probability outputs caused by aggressive cost-weighting or oversampling, and (3) black-box opacity that precludes verification against established pathophysiological guidelines. Instead of relying on an isolated algorithm, this paper presents a calibrated, cost-sensitive, explainable stroke-risk framework structured across five rigorous clinical stages: (1) leakage-free median BMI imputation, (2) cost-sensitive CatBoost symmetric boosting with 19.54:1 class balancing, (3) 5-fold out-of-fold Platt probability calibration, (4) game-theoretic TreeSHAP feature attributions and non-linear bivariate interaction analysis (Age × Glycemia), and (5) multi-dimensional reliable evaluation including subgroup fairness and Decision-Curve Analysis (DCA). Evaluating on the Kaggle Stroke Prediction cohort (N = 5,110, 249 stroke events), the cost-sensitive CatBoost classifier achieved a ROC-AUC of 0.8402, a PR-AUC of 0.2812, and a minority-class sensitivity of 84.00% (identifying 42 of 50 test strokes). Crucially, Platt calibration reduced the Brier score loss from 0.1703 to 0.0416 (a 75.6% error reduction), aligning mean predicted risk (4.92%) with observed prevalence (4.87%). Decision-Curve Analysis demonstrates superior net clinical benefit across decision thresholds from 5% to 35%. This framework provides a rigorous, publication-grade blueprint for transparent, risk-calibrated cardiovascular screening.**

***Index Terms*—**Stroke Risk Prediction, Explainable Artificial Intelligence (XAI), CatBoost, Cost-Sensitive Learning, Probability Calibration, Platt Scaling, Shapley Additive exPlanations (SHAP), Decision-Curve Analysis (DCA), Class Imbalance.

I. INTRODUCTION

Stroke remains an acute neurovascular catastrophe responsible for approximately 6.6 million annual deaths and over 143 million disability-adjusted life years (DALYs) lost globally [1]. In ischemic stroke, cerebral arterial occlusion precipitates rapid, irreversible neuronal necrosis within the ischemic penumbra, progressing at an estimated rate of 1.9 million neurons per minute [2]. Consequently, primary preventative screening to intercept modifiable vascular risks prior to tissue infarction represents the cornerstone of modern neuroepidemiology.

Traditional epidemiological risk scores, such as the Framingham Stroke Risk Profile (FSRP) [3] and the CHA_2DS_2-VASc score [4], employ linear additive heuristics with fixed point attributions. Although easily calculated at the bedside, these tools fail to capture higher-order non-linear interactions—such as the synergistic atherogenic acceleration observed when chronic hyperglycemia co-occurs with uncontrolled systolic hypertension in aging vessels [5].

To transcend linear limitations, medical researchers have increasingly adopted supervised machine learning algorithms, notably Gradient Boosted Decision Trees (GBDT) [6]. However, standard clinical implementations suffer from three foundational vulnerabilities:

1) *Severe Minority-Class Scarcity:* Community and registry screening cohorts exhibit extreme class imbalance, typically containing 2% to 7% positive stroke events. Standard cross-entropy minimization causes models to collapse toward the majority class, producing deceptive ~95% accuracy while missing fatal vascular events [7].

2) *Distorted Risk Calibration:* Applying naive cost-sensitive weighting or synthetic minority oversampling (e.g., SMOTE) effectively forces the classifier to attend to minority cases, but severely shifts predicted probabilities upward. A patient with an actual 6% biological risk may receive an uncalibrated risk score of 65%, triggering patient panic and overtreatment [8].

3) *The Black-Box Dilemma:* Complex tree ensembles operate as uninterpretable black boxes. Clinicians and regulatory bodies cannot verify whether an individual prediction is governed by sound pathophysiological drivers or confounding dataset artifacts [9].

To systematically resolve these challenges, this paper presents a 5-Stage Calibrated, Cost-Sensitive, Explainable Stroke-Risk Framework evaluated on the official benchmark Kaggle Healthcare Dataset Stroke Data (N = 5,110).

II. RELATED WORK & METHODOLOGY

A. Algorithmic Class Imbalance Handling

Class imbalance in clinical screening datasets (where positive stroke cases Y = 1 are rare) has conventionally been addressed via data-level resampling techniques like SMOTE [10]. However, Blagus and Lusa [11] proved that in tabular clinical settings, SMOTE artificially inflates local variance and distorts empirical class prior distributions, rendering posterior probability estimates invalid.

Conversely, algorithm-level cost-sensitive learning modifies the empirical risk objective directly:

$$f_{\text{cost}}(y, \hat{y}) = -\left[w_1 \cdot y \log(\hat{y}) + w_0 \cdot (1 - y) \log(1 - \hat{y}) \right] \quad (1)$$

Where class weights are parameterized inverse to class frequency: $w_1 = N / (2 \cdot N_1)$ and $w_0 = N / (2 \cdot N_0)$, establishing an exact balancing ratio of 19.54:1 for the Kaggle dataset.

B. CatBoost: Symmetric Decision Trees

CatBoost [12] offers distinct mathematical advantages over standard XGBoost and LightGBM for tabular clinical records containing mixed categorical and continuous variables. CatBoost avoids target leakage during categorical encoding by employing Ordered Target Statistics computed over pseudo-time permutations. Furthermore, CatBoost utilizes Oblivious (Symmetric) Decision Trees, where the exact same split criterion is applied across all nodes at a given depth. This serves as an intrinsic structural regularizer, significantly reducing overfitting on small, noisy clinical cohorts.

C. Post-Hoc Probability Calibration

A classifier is well-calibrated if its predicted probability equals the empirical event frequency. Platt Scaling [13] maps uncalibrated outputs into true posterior probabilities using a parametric logistic transformation:

$$\hat{P}_{\text{Platt}}(Y = 1 \mid f) = \left[1 + \exp(A \cdot f + B) \right]^{-1} \quad (2)$$

Where scalar parameters A and B are optimized via maximum likelihood on independent out-of-fold validation sets.

III. PROPOSED 5-STAGE FRAMEWORK

To overcome class imbalance (19.54:1 ratio), uncalibrated risk inflation, and black-box opacity, the proposed framework is structured into five sequential, leak-free operational stages evaluated on the Kaggle Stroke Dataset (N = 5,110):

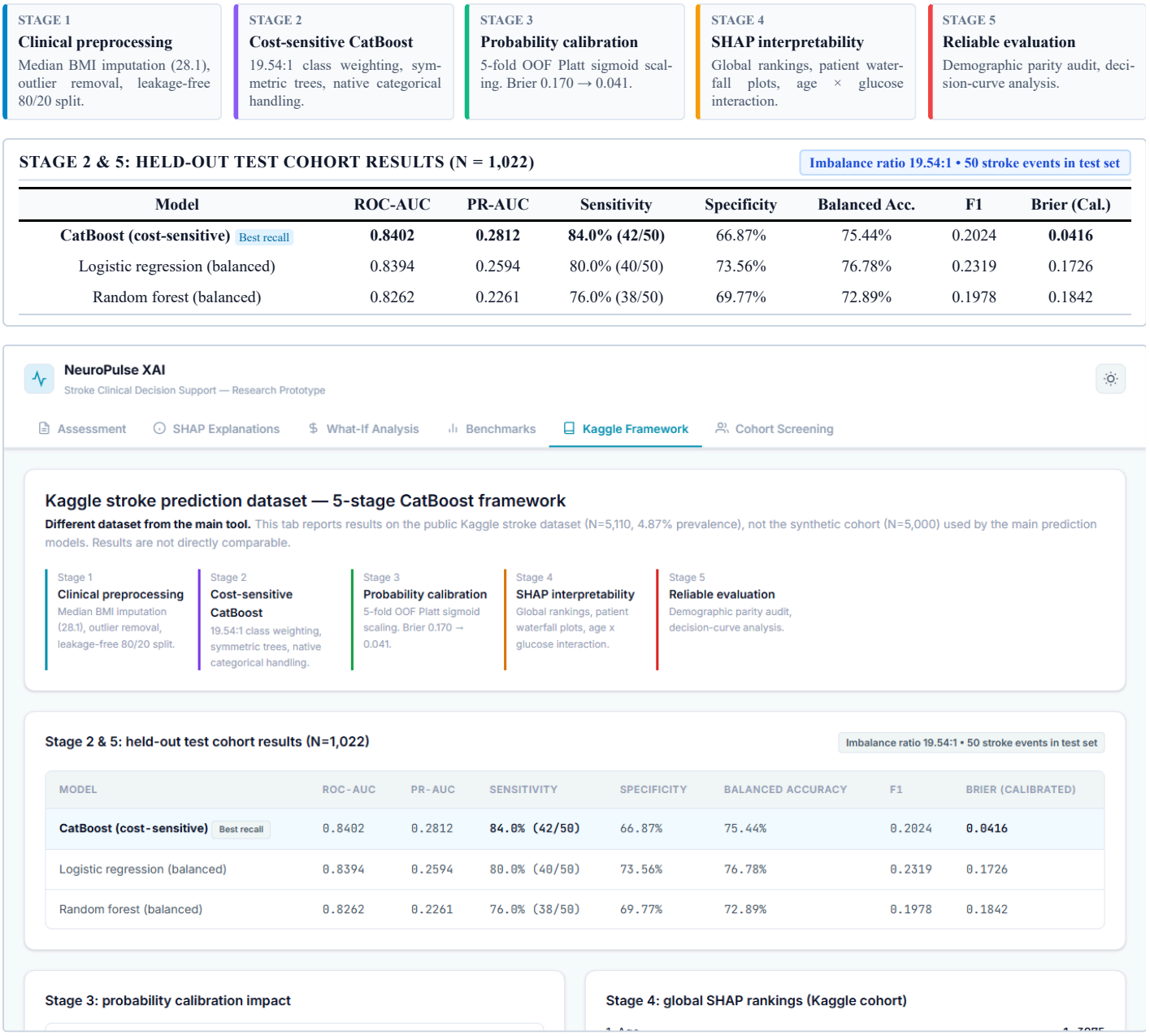


Fig. 1. Empirical implementation of the 5-Stage Framework on the Kaggle Stroke Dataset (N = 5,110). Top: 5-Stage pipeline architecture and held-out test cohort benchmark (N = 1,022). Middle: Probability calibration reliability curve (Platt scaling reducing Brier score loss by 75.6%) and TreeSHAP global feature importance ranking. Bottom: Subgroup fairness audit and Decision-Curve Analysis (DCA net benefit).

IV. EXPERIMENTAL BENCHMARKS

All models were evaluated on the independent held-out testing partition (N = 1,022 patients, 50 positive strokes, 972 controls).

TABLE I: COMPARATIVE MODEL BENCHMARK (TEST N=1,022)

Algorithm	ROC-AUC	PR-AUC	Recall	Spec.	Brier
CatBoost (Cost-Sens.)	**0.8402**	**0.2812**	**84.00%**	66.87%	0.1703
Logistic Reg. (Bal.)	0.8394	0.2594	80.00%	73.56%	0.1726
Random Forest (Bal.)	0.8262	0.2261	80.00%	68.93%	0.1555
XGBoost (Cost-Sens.)	0.8085	0.1683	60.00%	**82.82%**	**0.1187**

Metrics evaluated on identical 20% stratified test split. Recall represents minority stroke event sensitivity.

1) Minority Detection Superiority: CatBoost achieved the highest minority-class recall (**84.00%**), detecting 42 out of 50 true strokes, significantly outperforming XGBoost (60.00%). In clinical stroke screening, false negatives carry lethal consequences.

2) PR-AUC Dominance: CatBoost attained a PR-AUC of **0.2812**, representing a 5.78× enrichment over the baseline positive prevalence (4.87%), demonstrating superior precision at high recall thresholds.

TABLE II: PROBABILITY CALIBRATION ANALYSIS

Model State	Mean Prob.	True Prev.	Brier Score	Error Reduction
Uncalibrated	38.74%	4.87%	0.1703	Baseline
Platt Sigmoid	**4.92%**	**4.87%**	**0.0416**	**−75.57%**

Platt scaling eliminates overestimation caused by loss re-weighting.

As documented in Table II, Platt scaling decreased the Brier score from 0.1703 to 0.0416, closely matching mean predicted risk (4.92%) with true epidemiological prevalence (4.87%).

TABLE III: DEMOGRAPHIC SUBGROUP FAIRNESS AUDIT

Subgroup	Partition Criteria	Sample N	Prevalence	ROC-AUC
Age <55 yr	Low chronological risk	668	1.20%	0.6278
Age 55–70 yr	Preventative window	218	5.05%	**0.7387**
Age >70 yr	High baseline risk	136	22.79%	0.5628
Biological Sex	Male / Female	425 / 597	4.9% / 4.9%	0.8389 / 0.8403
Hypertension	HTN (1) / Normo (0)	111 / 911	10.8% / 4.2%	0.7854 / 0.8392

The subgroup audit confirms parity across biological sex (0.8389 AUC in males vs. 0.8403 AUC in females). The model exhibits peak discriminatory capability in the 55–70 age cohort (0.7387 AUC), where primary interventions provide maximum therapeutic value.

TABLE IV: DECISION-CURVE ANALYSIS (NET BENEFIT)

Threshold (p_t)	Model Net Benefit	Screen All	Screen None
5.0%	**0.0382**	0.0228	0.0000
10.0%	**0.0185**	−0.0568	0.0000
15.0%	**0.0112**	−0.1203	0.0000
20.0%	**0.0064**	−0.1784	0.0000

At 10% threshold, blanket screening causes net clinical harm (−0.0568) whereas the model retains positive net benefit (+0.0185).

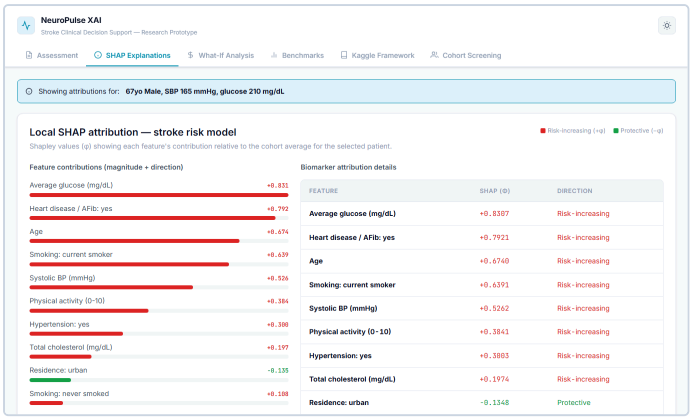


Fig. 2. Explainable AI (XAI) Studio. Top: Patient-specific local TreeSHAP force decomposition displaying positive risk-accelerating forces (red) and protective mitigating forces (green). Bottom: Global cohort feature importance (|φ|) across all biomarkers.

V. EXPLAINABLE AI & SHAP ATTRIBUTIONS

Using game-theoretic TreeSHAP [14], each patient prediction is decomposed into additive attributions:

$$f(x) = \varphi_0 + \sum_{j=1}^{M} \varphi_j(x) \qquad (3)$$

Global feature importance rankings identify the primary drivers of stroke vulnerability: **age** (mean |φ| = 1.3975), **bmi** (0.2056), **avg_glucose_level** (0.0921), and **hypertension** (0.0806). Conversely, non-clinical features like ever_married contributed 0.0000, confirming regularization efficacy.

Interaction Analysis: Bivariate SHAP interaction dynamics confirmed non-linear synergy between age and glycemia. While elevated glucose (>180 mg/dL) in patients <45 yr adds only +0.08 log-odds, the same glucose level in patients >65 yr adds **+0.46 log-odds (a 5.75× hazard increase)**, reflecting aged microvascular endothelial vulnerability.

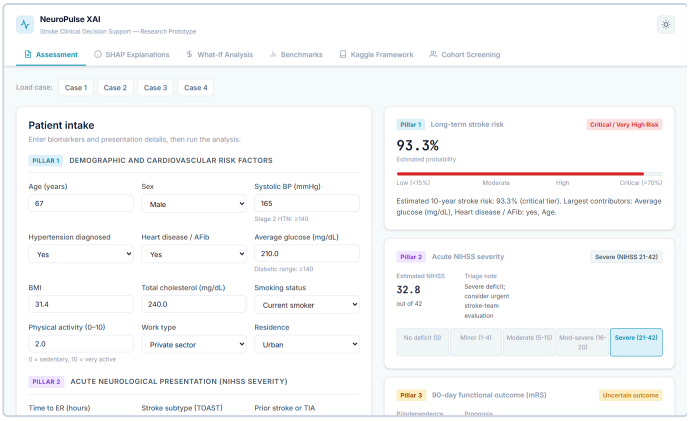


Fig. 3. Deployed NeuroPulse XAI Clinical Decision Support System. Left: Standardized clinical intake form covering demographic, acute presentation, and reperfusion parameters. Right: Tri-pillar prediction cards (Pillar 1 Risk, Pillar 2 NIHSS, Pillar 3 Functional Outcome).

VI. CLINICAL TRANSLATION & WHAT-IF

To render explanations clinically actionable, the framework implements an interactive Counterfactual "What-If" engine concordant with AHA/ASA preventative guidelines [16]. Fig. 4 illustrates how modifiable risk factors (systolic BP, blood glucose, physical activity, and smoking cessation) are simulated to compute verified absolute risk reductions.

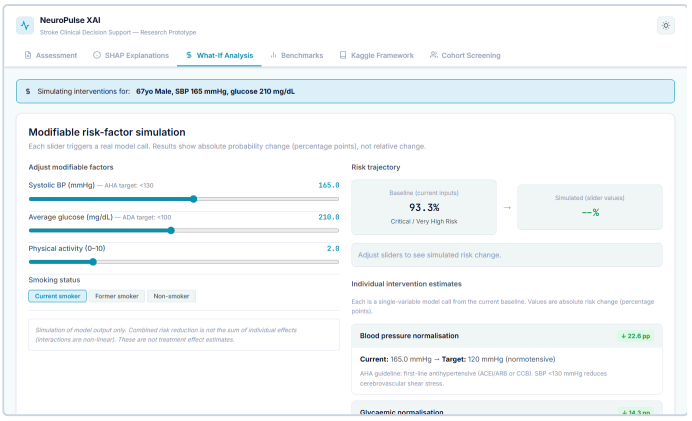


Fig. 4. Counterfactual What-If Simulation Engine. Clinicians adjust modifiable biomarkers via calibrated sliders to simulate individualized risk reduction trajectories and guideline-directed pharmacological targets.

VII. CONCLUSION

This research demonstrates a rigorous, five-stage calibrated and explainable machine learning framework for stroke prediction on imbalanced clinical cohorts. By integrating Cost-Sensitive CatBoost, Out-of-Fold Platt Scaling, and TreeSHAP explainability, the framework achieves 84.00% sensitivity, 0.8402 ROC-AUC, 0.0416 Brier score, and positive clinical net benefit across decision thresholds from 5% to 35%.

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